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Unpacking the Climate 'Hiatus': Why Global Warming Rises in Steps

Abstract

By mid-2015, global temperatures were breaking records again after a period when surface warming had slowed. Climate scientist Kevin Trenberth asked whether the hiatus was over. His analysis showed that a natural Pacific pattern, the Pacific Decadal Oscillation, controls where the ocean stores extra heat, making the global temperature record look less like a smooth ramp and more like a rising staircase.

Introduction

In 2015, signs of a heating planet were everywhere. Drought and wildfires spread across western North America, the Pacific was unusually active with typhoons, and U.S. temperatures ran 1.7°F above the 20th-century average. Global mean surface temperature had been rising fairly steadily: every decade since the 1960s was warmer than the one before. Yet there were two periods when warming slowed sharply, from about 1943 to 1975 and from 1999 to 2013. The second became known as the hiatus. With 2014 setting a new record and 2015 looking likely to beat it, Trenberth examined what caused these slowdowns and why warming seemed to be taking another step up.

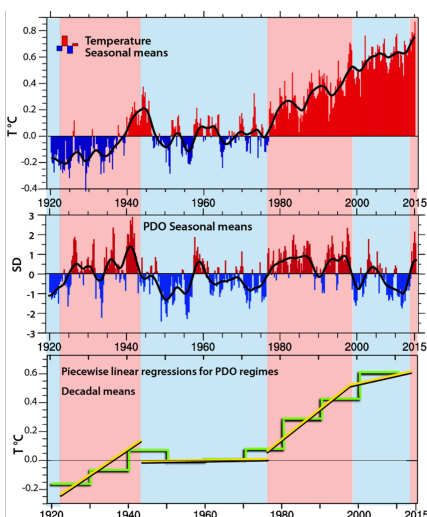
How Do Scientists Study This?



FROM THE ARTICLE

"Are we in for more heat?" Drought-cracked ground. Photo: suburbanbloke/flickr, CC BY-SA.

Trenberth analyzed NOAA's record of seasonal global mean surface temperatures since 1920 alongside the Pacific Decadal Oscillation (PDO), a decades-long pattern of Pacific Ocean temperatures. He divided the record into periods when the PDO was in its positive or negative phase and calculated the warming rate within each phase. He also examined how the PDO changes trade winds, ocean currents, and where heat is stored, comparing the upper 300 meters of the ocean with deeper layers. Finally, he considered outside influences, such as air pollution particles from industry, and compared observed warming rates with climate model projections.

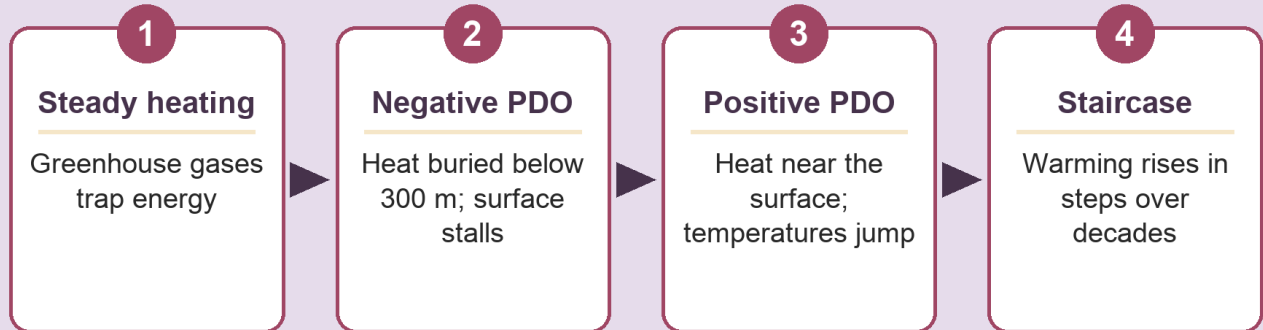


FROM THE ARTICLE

Top: seasonal global temperatures since 1920. Middle: the PDO, with positive (pink) and negative (blue) phases. Bottom: decadal averages rising in steps. Kevin Trenberth, data from NOAA.

Explore the Evidence

How the PDO shapes surface warming



Plot the record

Global temperatures since 1920 were compared with the phases of the Pacific Decadal Oscillation.

Measure each phase

Warming rates were calculated separately for positive and negative PDO periods.

Follow the heat

Scientists compared heat stored in the upper 300 m of ocean with heat buried deeper.

What Did They Discover?

Trenberth found that the PDO is the main driver of slowdowns in global surface warming. During its positive phase, more heat stays in the upper 300 meters of the ocean, where it warms the surface, so global temperature rises. During its negative phase, more heat is buried below 300 meters, and surface temperatures stagnate, even though Earth's total energy imbalance is largely unchanged. The 1998 record came from the biggest El Niño on record, and a strong new El Niño in 2015 was driving recent warmth. The earlier slowdown was also affected by post-war air pollution, which reflected sunlight until clean air laws such as the U.S. Clean Air Act of 1970 reduced it.

The hiatus by the numbers

1999–2013 the most recent hiatus in surface warming

300 m depth dividing surface-warming heat from buried heat

1.7°F U.S. temperatures above the 20th-century average in 2015

Climate models suggest the human signal emerged from natural variability around the 1970s. Human-caused warming is relentless, even when natural cycles mask it for a while.

Takeaways

A rising staircase

Natural cycles plus steady heating make warming rise in steps, not a smooth ramp.

The Pacific decides

The PDO controls whether extra heat warms the surface or sinks deeper.

Heat never left

During the hiatus, the planet kept gaining energy; it was stored in the deep ocean.

Key Terms

Global mean surface temperature

The average temperature of the air near Earth's surface, combining land and ocean measurements worldwide.

Pacific Decadal Oscillation (PDO)

A natural pattern of Pacific Ocean temperatures that switches between positive and negative phases over decades.

Natural variability

Ups and downs in climate caused by natural interactions among the ocean, atmosphere, land, and ice.

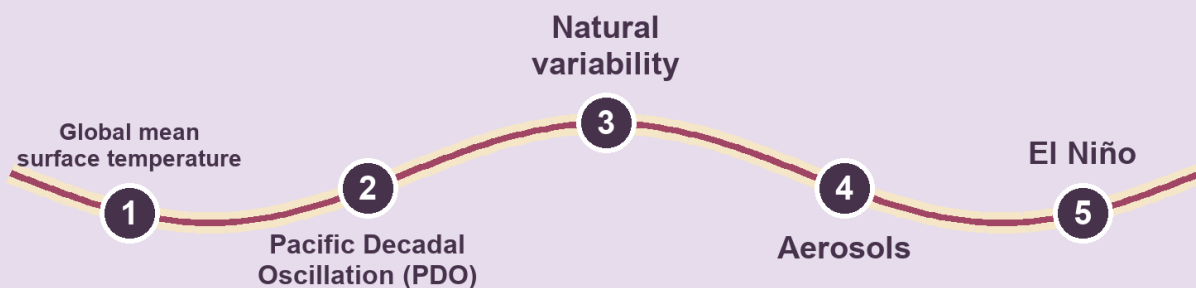
Aerosols

Tiny particles in the air, often from pollution, that can reflect sunlight and temporarily cool the planet.

El Niño

A natural warming of the tropical Pacific Ocean that raises global temperatures and changes weather worldwide.

Vocabulary Path · Unpacking the Climate 'Hiatus'



Activity

Build the Climate Staircase

Students model how a steady trend plus natural cycles produces a staircase pattern in temperature data.

Materials: Graph paper; Two colored pencils; Calculator or spreadsheet

1. Plot a steady warming trend: add 0.2 units every 10 years for 100 years.
2. Add a simple cycle: for 30 years subtract a little each decade, then for 30 years add a little extra, and repeat.
3. Plot the combined line in a second color. Where do you see steps and flat spots?
4. Label the flat spots as hiatus periods and the steep parts as positive PDO phases.
5. Explain in writing why a flat period in the data does not mean warming has stopped.

References

Trenberth, K. (2015). Is the global warming 'hiatus' over? The Conversation.
<https://doi.org/10.64628/AAI.pdunhja7d>

Trenberth, K. E. (2015). Has there been a global warming hiatus? Paper described in the article above.